

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a differentiable function, the function $\nabla f(x) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is called the gradient of f and is defined as

$$\nabla f(x_1, x_2, x_3) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \frac{\partial f}{\partial x_3} \end{pmatrix}.$$

Therefore, the gradient of the function is

$$\nabla f(x_1, x_2, x_3) = \begin{pmatrix} e^{x_1} + 2x_1x_3 - x_2x_3 \\ -x_1x_3 \\ x_1^2 - x_1x_2 \end{pmatrix}.$$

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function. Consider $x \in \mathbb{R}^n$ and $d \in \mathbb{R}^n$. The directional derivative of f at x in the direction d is given by

$$\lim_{\alpha \rightarrow 0} \frac{f(x + \alpha d) - f(x)}{\alpha},$$

if the limit exists. Here, we have

$$\begin{aligned} f(x + \alpha d) - f(x) &= e^{x_1 + \alpha d_1} + (x_1 + \alpha d_1)^2(x_3 + \alpha d_3) \\ &\quad - (x_1 + \alpha d_1)(x_2 + \alpha d_2)(x_3 + \alpha d_3) \\ &\quad - (e^{x_1} + x_1^2x_3 - x_1x_2x_3) \\ &= e^{x_1 + \alpha d_1} - e^{x_1} + \alpha x_1^2 d_3 + \alpha^2 x_3 d_1^2 + \alpha^3 d_1^2 d_3 + 2\alpha x_1 x_3 d_1 \\ &\quad + 2\alpha^2 x_1 d_1 d_3 - \alpha x_1 x_2 d_3 - \alpha x_1 x_3 d_2 - \alpha^2 x_1 d_2 d_3 \\ &\quad - \alpha x_2 x_3 d_1 - \alpha^2 x_2 d_1 d_3 - \alpha^2 x_3 d_1 d_2 - \alpha^3 d_1 d_2 d_3. \end{aligned}$$

Therefore,

$$\begin{aligned} \lim_{\alpha \rightarrow 0} \frac{f(x + \alpha d) - f(x)}{\alpha} &= \lim_{\alpha \rightarrow 0} \frac{e^{x_1 + \alpha d_1} - e^{x_1}}{\alpha} \\ &\quad + x_1^2 d_3 + 2x_1 x_3 d_1 - x_1 x_2 d_3 - x_1 x_3 d_2 - x_2 x_3 d_1 \\ &= d_1 e^{x_1} + x_1^2 d_3 + 2x_1 x_3 d_1 - x_1 x_2 d_3 - x_1 x_3 d_2 - x_2 x_3 d_1 \\ &= d_1 (e^{x_1} + 2x_1 x_3 - x_2 x_3) - x_1 x_3 d_2 + (x_1^2 - x_1 x_2) d_3. \end{aligned}$$

In addition, when the gradient exists, the directional derivative is the inner product between the gradient of f and the direction d , that is,

$$\nabla f(x)^T d.$$

Using this formula, the directional derivative is

$$\begin{aligned} (d_1 \ d_2 \ d_3) \nabla f(x_1, x_2, x_3) &= \\ d_1 (e^{x_1} + 2x_1 x_3 - x_2 x_3) &- d_2 x_1 x_3 + d_3 (x_1^2 - x_1 x_2). \end{aligned}$$

You can verify that it is the same result as above.